EXPERIMENT – 5

```
clc;
clear;
close all;

% Microwave Directional Coupler and Two-Way Power Divider
% Frequency range: 8 GHz to 12 GHz

f = linspace(8,12,201);
x = (f-10).^2;

% Directional Coupler S-parameters (approximate values)
S11 = -25 - 0.75*x;
S21 = -0.5 - 0.075*x;
S31 = -10.5 - 0.375*x;
S41 = -30 + 1.0*x;

% Coupler performance parameters
Coupling = 10 + 0.5*x;
Isolation = 30 - 2.5*x;
InsertionLoss = 0.5 + 0.1*x;
Directivity = 20 - 1.5*x;

% Two-way Power Divider S-parameters
D_S11 = -25 - 0.75*x;
D_S21 = -3.0 - 0.1*x;
D_S31 = -3.0 - 0.1*x;
D_Isolation = -30*ones(size(f));
PowerRatio = ones(size(f));

figure('Color','w','Position',[50 50 1400 650]);
```

```
% 1. Coupler S11
subplot(3,4,1);
plot(f,S11,'b','LineWidth',1.8);
title('Coupler S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;

% 2. Coupler S21
subplot(3,4,2);
plot(f,S21,'b','LineWidth',1.8);
title('Coupler S21');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;

% 3. Coupler S31
subplot(3,4,3);
plot(f,S31,'b','LineWidth',1.8);
title('Coupler S31');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;

% 4. Coupler S41 - Isolated Port
subplot(3,4,4);
plot(f,S41,'b','LineWidth',1.8);
title('Coupler S41 - Isolated Port');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;
```

```
% 5. Coupling
subplot(3,4,5);
plot(f,Coupling,'b','LineWidth',1.8);
hold on;
yline(10,'r--','10 dB Target','LineWidth',1.2);
title('Coupling');
xlabel('Frequency (GHz)');
ylabel('Coupling (dB)');
grid on;

% 6. Isolation
subplot(3,4,6);
plot(f,Isolation,'b','LineWidth',1.8);
hold on;
yline(25,'r--','25 dB Target','LineWidth',1.2);
title('Isolation');
xlabel('Frequency (GHz)');
ylabel('Isolation (dB)');
grid on;

% 7. Insertion Loss
subplot(3,4,7);
plot(f,InsertionLoss,'b','LineWidth',1.8);
hold on;
yline(0.5,'r--','0.5 dB Limit','LineWidth',1.2);
title('Insertion Loss');
xlabel('Frequency (GHz)');
ylabel('Loss (dB)');
grid on;

% 8. Directivity
```

```
subplot(3,4,8);
plot(f,Directivity,'b','LineWidth',1.8);
hold on;
yline(15,'r--','15 dB Target','LineWidth',1.2);
title('Directivity');
xlabel('Frequency (GHz)');
ylabel('Directivity (dB)');
grid on;

% 9. Divider S11
subplot(3,4,9);
plot(f,D_S11,'b','LineWidth',1.8);
title('Divider S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;

% 10. Divider output S21 and S31
subplot(3,4,10);
plot(f,D_S21,'b','LineWidth',1.8);
hold on;
plot(f,D_S31,'r--','LineWidth',1.8);
title('Divider Output S21 and S31');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
legend('S21','S31','Location','best');
grid on;

% 11. Divider Output Isolation
subplot(3,4,11);
plot(f,D_Isolation,'b','LineWidth',1.8);
title('Divider Output Isolation');
```

```
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
grid on;

% 12. Power Division Ratio
subplot(3,4,12);
plot(f,PowerRatio,'b','LineWidth',1.8);
hold on;
yline(1,'r--','Ideal 1:1','LineWidth',1.2);
title('Power Division Ratio');
xlabel('Frequency (GHz)');
ylabel('P2 / P3');
grid on;

sgtitle('Microwave Directional Coupler and Two-Way Power Divider');
```

Microwave Directional Coupler and Two-Way Power Divider





